Chapter 1

How Plants Make Their Food

All living things need food. Animals eat plants or other animals. But plants do not eat anything. Plants make their own food, and they make it in their leaves. That is why green leaves are called the kitchens of plants.

Leaves are green because they have a green substance called chlorophyll. Chlorophyll helps the leaves catch sunlight.

What a plant needs to make food

1. Sunlight - the leaves catch it with the help of chlorophyll.
2. Water - the roots take in water from the soil. The water travels up through the stem to the leaves.
3. Air - the leaves take in a gas called carbon dioxide from the air through tiny holes called stomata. Stomata are so small that we cannot see them without a microscope.

Using the energy of sunlight, the leaves join water and carbon dioxide to make food. This process is called photosynthesis. "Photo" means light and "synthesis" means putting together. While making food, plants give out oxygen into the air. Oxygen is the gas that we and all animals need to breathe.

Where plants store food

Plants make more food than they need at once. They store the extra food in different parts. We eat many of these parts.

- Potato is a stem that stores food under the ground.
- Carrot and radish are roots that store food.
- Wheat, rice and gram are seeds that store food.
- Mango and banana are fruits that store food.

Much of this stored food is starch.

Activity: Test for starch

Take a slice of potato. Put two drops of iodine solution on it with the help of your teacher. The place where the iodine falls turns blue-black. This shows that the potato contains starch.

Activity: Plants need sunlight

Keep one potted plant in a dark cupboard and another in the sun for a week, and water both. The leaves of the plant in the cupboard turn pale and yellow, because without sunlight it cannot make food.

Let's think

1. Why are leaves called the kitchens of plants?
2. Name the three things a plant needs to make food.
3. What gas do plants give out during photosynthesis?

New words

chlorophyll - the green substance in leaves
stomata - tiny holes in leaves through which air goes in and out